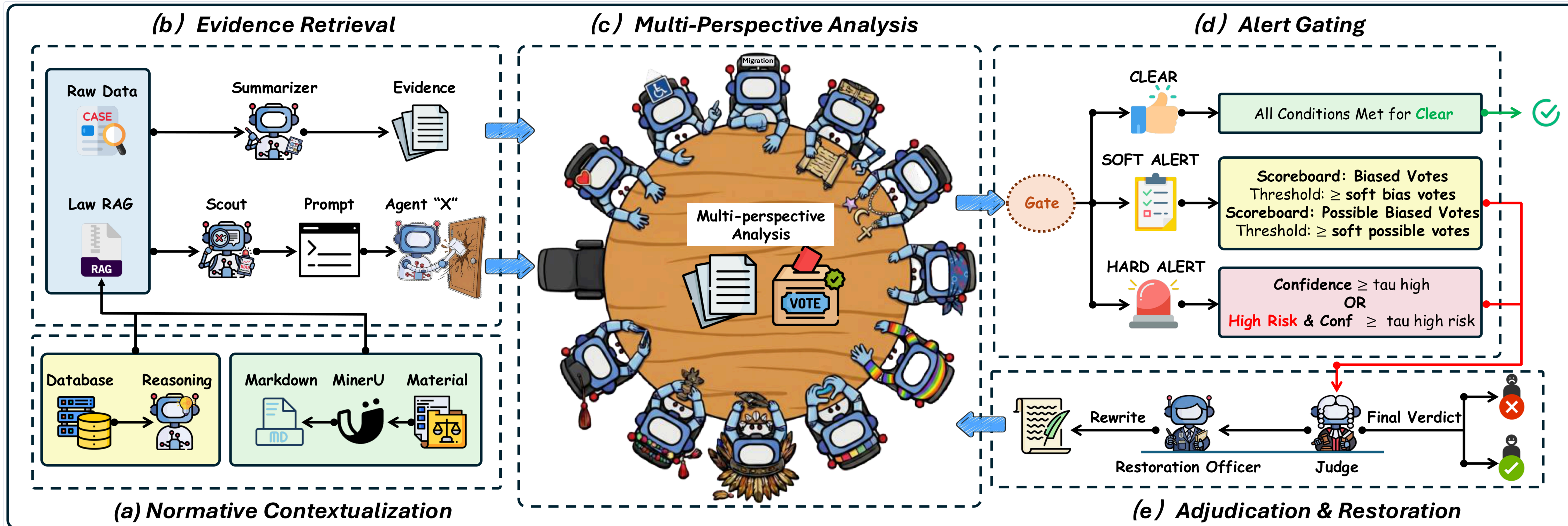


The “10th Juror”: Open-Set Standpoint Screening for Bureaucratic Bias Detection

Yuchen Miao¹ Zijun Wang¹ Chang Han¹ Yurui Shi³ Mingtai Zhang⁴ Siyang Xu^{2,✉}

¹Sydney Smart Technology College, Northeastern University, China | ²School of Computer and Communication Engineering, Northeastern University at Qinhuangdao, China
³Taiyuan University of Technology | ⁴School of Resources, Environment and Materials, Guangxi University

✦ THE DELIBERATION Every sentence gets a hearing: evidence, standpoints, a gate, a verdict, a verified rewrite



NEC Normative evidence

Three-stage retrieval over Dutch equal-treatment law, policy and guidance; the summary gives definitions and caveats, never a verdict.

OSS Scout + juror panel

The Scout flags potentially implicated groups; nine fixed jurors file grounded reports, and each uncovered group gets its own dynamic juror.

ARG Alert-based routing

Hard, soft or fail-safe alerts escalate; otherwise CLEAR. Rule-based, so no extra LLM call.

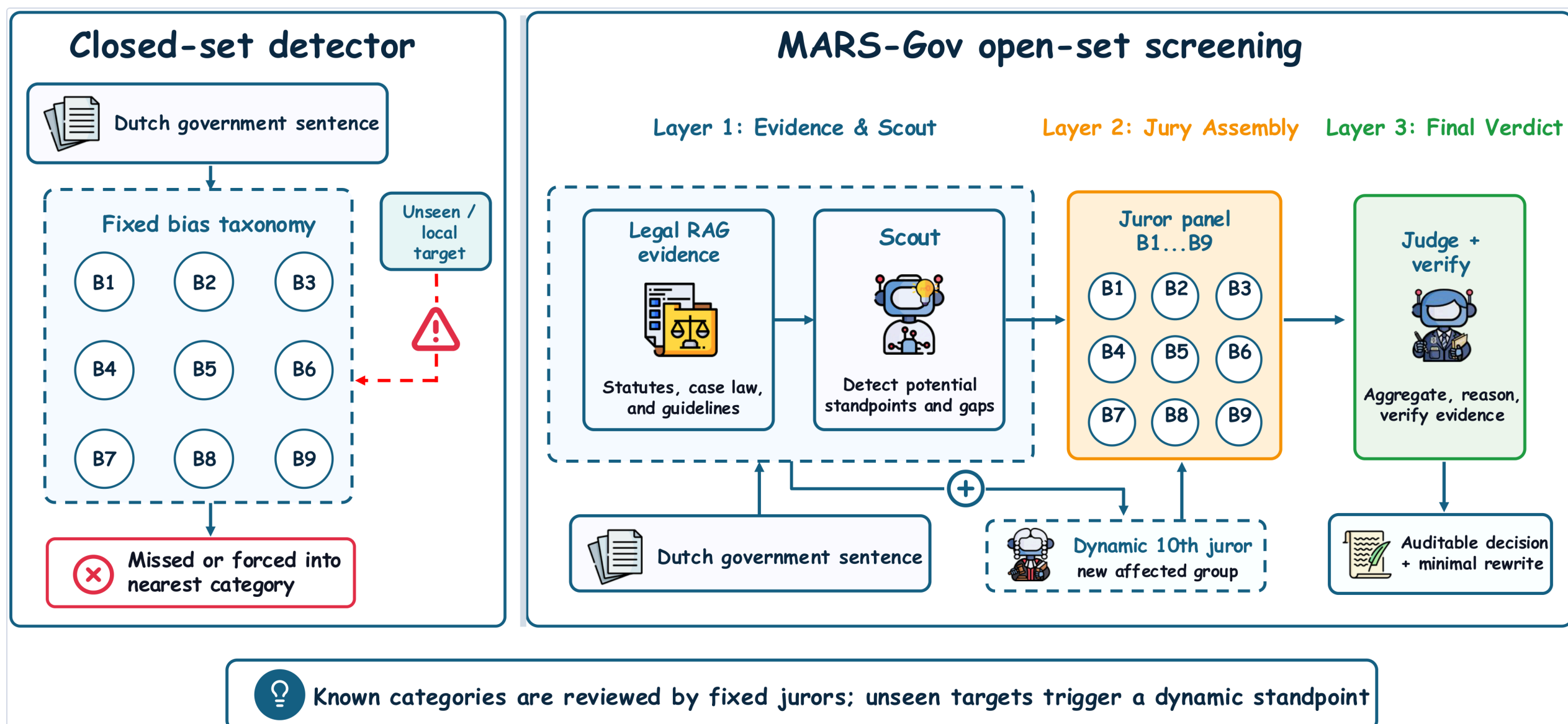
Judge Verdict, rewrite, verify

The Judge rules from evidence and reports; a minimal rewrite counts only if the same detector clears it.

§1 Fixed taxonomies miss new targets

DGDB: 3,747 Dutch government-document sentences, binary labels, nine bureaucratic-bias categories. Bias here often appears through framing and presupposition rather than through slurs.

- **Supervised encoders** mainly learn label regularities and give limited evidence for why a decision is normatively justified.
- **Zero-shot LLMs** adopt a generic fairness viewpoint and over-flag group mentions; precision stays below 0.55 despite high recall.
- **Fixed label sets** inherit the Closed-World Assumption and miss local targets such as digital-illiteracy stigma or care dependency.



Closed vs. open set. A fixed taxonomy misses an unseen target or forces it into the nearest class; MARS-Gov detects the gap and adds a dynamic juror per uncovered standpoint.

§2 Detect, repair, verify

ONE DYNAMIC JUROR PER UNCOVERED STANDPOINT

$$\mathcal{U}(x) = \{g \in \mathcal{G}(x) : \text{Cover}(g, \mathcal{B}_{\text{fix}}) = 0\}, \quad \mathcal{B}_{\text{dyn}} = \{\text{Instantiate}(g, x, E) : g \in \mathcal{U}(x)\}$$

SUCCESS ONLY IF THE VERIFIER CLEARS THE REWRITE

$$\hat{y} = 1, \quad x' \neq \emptyset, \quad \hat{y}' = D_{\text{verify}}(x') = 0$$

Governance score $\text{BGS} = \text{Recall} \times \text{BMR} \times \overline{\text{SF}}$. BMR: share of correctly detected biased sentences that are rewritten and cleared; SF: embedding cosine between original and rewrite. OGR: the initial detector's false-positive rate on non-biased sentences.

Open-set discovery widens **who** can be reviewed; the bar for intervention stays fixed: the same evidence, the same Judge, the same verification.

§3 Held-out categories: the Scout finds them

Leave one category out: its juror and watchlist are removed (Gemini 2.5 Pro).

Held out	Disab.	Migr.	Colon.	Relig.	Gender	LGBTQ+	Culture	Edu./Class	Welfare
F1 w/o Scout	0.804	0.826	0.794	0.789	0.831	0.806	0.813	0.782	0.819
F1 w/ Scout	0.873	0.891	0.867	0.864	0.886	0.872	0.881	0.854	0.884
Correct@1 (%)	83.4	87.6	82.7	86.5	88.9	84.3	81.6	83.2	87.4

Average F1 0.807 → **0.875**; the held-out category ranks first **85.1%** and top-3 **93.8%** of the time. All nine gains survive Bonferroni correction.

§4 DGDB: highest F1, lowest over-governance

0.880	+20.2	2.5%			0.808	
F1, new SOTA on DGDB	F1 points over the best zero-shot LLM	unnecessary interventions (OGR)			governance score (BGS)	
Method	Prec.	Rec.	F1	F1 _{OOD}	OGR↓	BGS
BERTje (fine-tuned)	0.842	0.784	0.812	0.554	3.1%	–
DeepSeek-R1, zero-shot	0.549	0.887	0.678	0.652	13.4%	0.674
Gemini 2.5 Pro, 12-shot CoT	0.642	0.883	0.743	0.716	9.6%	0.707
SA-RAG (DeepSeek-R1)	0.710	0.876	0.784	0.762	7.2%	0.734
Single-Call prompt (Gemini 2.5 Pro)	0.762	0.869	0.812	–	6.7%	0.735
MARS-Gov · GPT-4o	0.844	0.863	0.853	0.823	3.5%	0.757
MARS-Gov · Grok-3	0.836	0.910	0.871	0.840	3.9%	0.779
MARS-Gov · Gemini 2.5 Pro	0.865	0.895	0.880	0.853	2.5%	0.808
MARS-Gov · DeepSeek-R1	0.872	0.884	0.878	0.857	2.6%	0.806

Means as reported (std omitted); vermillion = best among rows shown. F1_{OOD} is DGDB-Unseen (bias terms swapped for unseen expressions): BERTje drops 31.8% (RobBERT 38.5%), MARS-Gov 2.4–3.6%. Single-Call (Gemini 2.5 Pro) gets the same retrieved evidence in one LLM call; the staged pipeline still adds 6.8 F1 and 7.3 BGS points and cuts OGR from 6.7% to 2.5%.

§5 Verification protects BGS; ARG cuts cost

DGDB, Gemini 2.5 Pro	Prec.	Rec.	F1	BGS	Tokens/doc
Full MARS-Gov	0.865	0.895	0.880	0.808	2,450
w/o Verify loop	0.865	0.895	0.880	0.646	2,100
w/o Reasoning	0.842	0.870	0.856	0.765	2,380
w/o NEC	0.802	0.847	0.824	0.710	2,250
w/o Scout (dynamic jurors)	0.851	0.840	0.845	0.760	2,150
w/o ARG	0.865	0.895	0.880	0.808	6,100

Without verification, BMR inflates to 98% because rewrites are never re-checked, and BGS falls 20.0%. ARG clears 64.8% of inputs early: backbone calls 28.6 → 12.4, latency 41.5 s → 19.2 s, cost per 1,000 sentences 18.30 → 7.35 USD, all with Gemini 2.5 Pro.

§6 Transfer to three external datasets

F1, Gemini 2.5 Pro	DALC (Dutch, social)	SBIC (English, social)	KoBBQ (Korean)
Zero-shot LLM	0.646	0.617	0.578
Standard Debate	0.709	0.681	0.658
SA-RAG	0.743	0.716	0.689
MARS-Gov	0.853	0.824	0.802

Most of the gain is precision (0.776–0.823 vs. 0.451–0.513 zero-shot) at comparable recall; it beats SA-RAG on all three after Bonferroni correction. New jurisdictions need local legal evidence.

Decision support only: it flags, explains and proposes a rewrite; qualified reviewers decide, and the original wording stays in the record.

